Elice Priyadarshini

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EDUCATION

University of Wisconsin-Madison (GPA – 3.8) Madison, WI
Master of Science in Electrical and Computer Engineering *Aug 2024 – Dec 2025*
Major: Machine Learning and Signal Processing
Relevant Coursework: Machine Learning, Artificial Neural Networks, High-Performance Computing with CUDA, Advanced Image Processing, Software Engineering

Cummins College of Engineering for Women, Pune, India (GPA – 9.12) Pune, India
Bachelor of Technology in Electronics and Telecommunications Engineering *Aug 2016- May 2020*
Relevant Coursework: Advanced Digital Signal Processing, Image Processing, Big Data, Computer Vision, Remote Sensing

Projects

Fine-Tuning LLMs with Chain of Thought Reasoning | *Lee Lab, University of Wisconsin-Madison*
Independent Study Researcher Sept 2024 – Dec 2024

- Fine-tuned an open-source LLM using **Chain of Thought** reasoning techniques to enhance interpretability.
- Developed a **process reward model** leveraging **reinforcement learning** over synthetic datasets across multiple domains.
- Designed and executed experiments, achieving **75.2% accuracy on GSM8K dataset**.

BridgeML | Open-Source ML Knowledge Repository
Founder & Author Dec 2024 – Present

- Created an **open-source repository** to document and share **personal ML notes, research insights, and discussions on academic papers**.
- Writing structured explanations on **ML algorithms, theoretical foundations, and practical applications** to make ML concepts easier to understand.
- Curating and analyzing **key research papers**, summarizing complex ideas, and discussing their real-world implications.
- Aiming to expand the repository into a collaborative learning space for **structured ML education**.

Collatz Conjecture – Mathematical Exploration using Large Language Models
Independent Researcher Jan 2025 – Present

- Investigated the **Collatz Conjecture** through **computational analysis** and **machine learning techniques**.
- Developed a **neural network model** to recognize patterns in sequences and predict iterative behavior.
- Exploring **LLM-driven approaches** to derive novel insights into the problem and assess heuristic strategies.

EXPERIENCE

Siemens Digital Industries Software | Pune, India
Software Engineer, September 2020 – July 2024

- **Built and shipped 17 applications** across various industries, leveraging **Siemens Teamcenter** for industry-specific **customizations**.
- Developed an **AI-driven recommendation system** using **regression models** for sustainable product BOM (bill of materials) design.
- Built a **chatbot widget leveraging the TAPAS model** for locally hosted **SQL-based NLP queries** to enhance data accessibility.
- Created **low-code platform widgets**, including **PDF annotation tools, process automation tools, and graph visualizations**.

SKILLS

Programming Languages: Python, C, C++, Julia, SQL, Node.js, React.js, JavaScript, TypeScript, Tailwind CSS
Frameworks and Tools: React.js, MATLAB, Mendix, TensorFlow, PyTorch, LangChain, scikit-learn, Keras
Machine Learning: Regression Models, Neural Networks, Natural Language Processing (NLP), Large Language Models (LLMs), Reinforcement Learning, Multimodal Models, GPU Acceleration (CUDA)
Software Engineering: Data Structures, Algorithms, Design Patterns, System Design, Clean Code Practices